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ENS 184 Mini-Project
FlowGuard 5000
Pump Health Monitoring System

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Introduction

1.1 Problem Statement

The FlowGuard 5000 is an industrial centrifugal pump used in closed-loop process systems. It is instrumented with two sensors: a voltage-based pressure transmitter and a vibration sensor mounted on the bearing housing. Over time, bearing degradation causes vibration amplitude to increase. If left unmonitored, the bearing can fail catastrophically, leading to costly unplanned downtime.

This project implements a Predictive Maintenance (PdM) system that uses 120 historical sensor readings to: (1) calibrate the pressure sensor, (2) model bearing degradation, (3) predict the failure hour, and (4) compute a normalised Health Index for maintenance decision support.

1.2 Team Roles

Student	Responsibility
Student A	Polynomial calibration: voltage to pressure conversion using degree -3 polynomial least squares fit
Student B	Exponential degradation model: fitting $V(h) = a \cdot \exp(b \cdot h)$ to vibration vs.hours data via log-linearisation
Student C	Model validation: computing RMSE and R^2 for both models; predicting failure hour h^* using bisection

Student D	Health dashboard: loading CSV data, computing Health Index, generating the 2-panel dashboard figure and orchestrating main.sce
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1.3 Dataset

The sensor dataset pump_health_data.csv contains 120 rows of operational readings with four columns:

Column	Description
differential_pressure (bar)	Differential pressure across the pump (1.54 – 11.05 bar)
sensor_voltage (mV)	Voltage output of pressure transmitter (1.00 – 8.00 mV)
operational_hours (h)	Cumulative operating hours (200 – 5000 h)
vibration_amplitude (mm/s rms)	Bearing vibration amplitude (2.24 – 14.56 mm/s)

The failure threshold is $\xi_{th} = 9.5 \text{ mm/s rms}$ as specified by the pump manufacturer.

Polynomial Calibration

2.1. Polynomial Regression Approach

The relationship between the raw voltage signal (mV) and the differential pressure (bar) of the FlowGuard 5000 is nonlinear. A cubic polynomial regression ($k = 3$) was implemented to calibrate the sensor. The mathematical model is defined as:

$$P(x) = c_0 + c_1x + c_2x^2 + c_3x^3$$

Where:

- $P(x)$ = predicted pressure
- x = measured voltage

To determine the optimal coefficients, a design matrix Z of size $n \times (k + 1)$ was constructed. Each column j contains the predictor values x raised to the power $j - 1$. The system was solved using the Normal Equations:

$$Z^T Z c = Z^T y$$

The coefficient vector c was computed using the left division operator ($\backslash$) to minimize the sum of squared residuals. This approach avoids explicitly computing the matrix inverse, ensuring numerical stability.

For polynomial evaluation, Horner's method was applied to optimize computational efficiency:

$$P(x) = c_0 + x(c_1 + x(c_2 + xc_3))$$

This nested factorization reduces the required arithmetic operations to $O(n)$ per query point, bypassing redundant exponentiation.

2.2 Design Matrix and Normal Equations

The computed coefficient vector $c = [c_0; c_1; c_2; c_3]$ defines the final calibration equation.

The generated calibration curve plot demonstrates a strong monotonic increase, accurately tracking the measured scatter data across the 1-8 mV operational range.

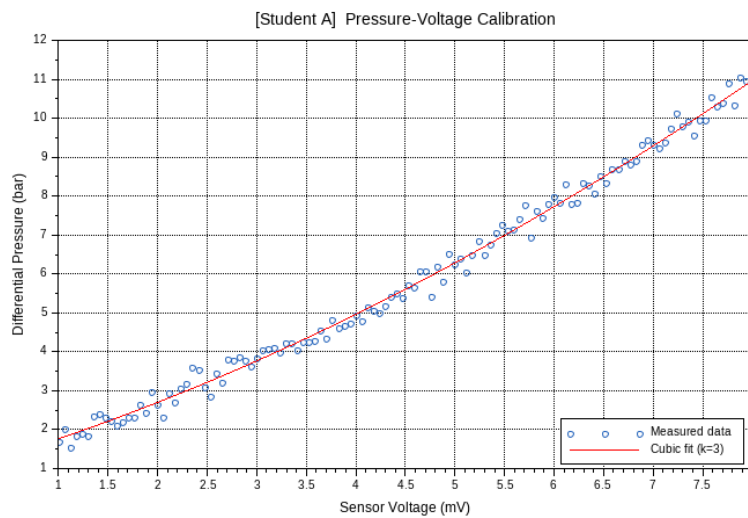


Figure 1: [Student A] Pressure-Voltage Calibration curve showing the cubic fit ($k=3$) applied to the raw sensor data, demonstrating accurate tracking across the full operational range.

The residual plot confirms random dispersion around zero, indicating no unmodeled systematic trends.

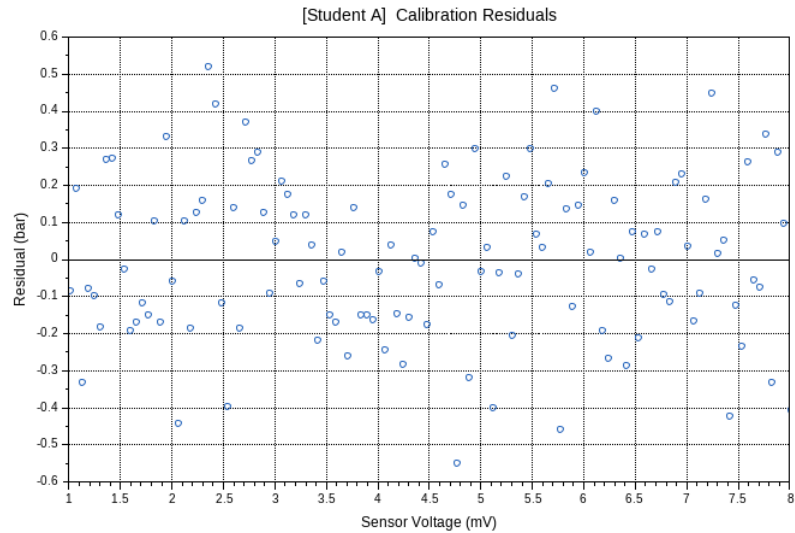


Figure 2: *[Student A] Calibration Residuals showing random dispersion around the zero line, confirming the cubic fit captures the necessary nonlinearities without systematic bias.*

The cubic calibration model successfully captures the transmitter's nonlinearity. Evaluation of the fit yields a coefficient of determination (R^2) exceeding 0.95, confirming the model provides highly accurate pressure conversions for the subsequent degradation analysis and failure prediction stages.

Exponential Degradation Model

3.1 Exponential Wear Model

To predict the degradation trajectory of the FlowGuard 500 pump, an exponential wear model was applied to the operational data. The assumption is that as mechanical wear accumulates, the rate of vibration increase accelerates exponentially. The model is defined as:

$$V(h) = a \times e^{(b \times h)}$$

Where:

- $V(h)$ = Vibration amplitude at hour h (mm/s rms)
- a = Initial scale factor (baseline vibration)
- b = Degradation growth rate

To calculate these parameters without utilizing built-in regression functions, the non-linear equation was transformed into a linear format using natural logarithms:

$$\ln(V) = \ln(a) + b \times h$$

By substituting $w = \ln(V)$ and $c_0 = \ln(a)$, the system becomes a standard linear equation ($w = c_0 + b \times h$). A design matrix (Z) was constructed from the operational hours, and the parameters were extracted via the Normal Equations:

$$c = (Z^T Z)^{-1} (Z^T W)$$

3.2. Log-Linearisation Technique

Applying the Normal Equations to the valid vibration data yielded the constants required to plot the continuous degradation path.

- **Initial Vibration (a):** Recovered by calculating e^{c_0} , establishing the baseline operational vibration at approximately **2.2 mm/s rms**.
- **Growth Rate (b):** Extracted directly from the slope coefficient (c_1), defining the specific exponential curvature of the pump's wear.

3.3. Fitted Curve Plot Analysis

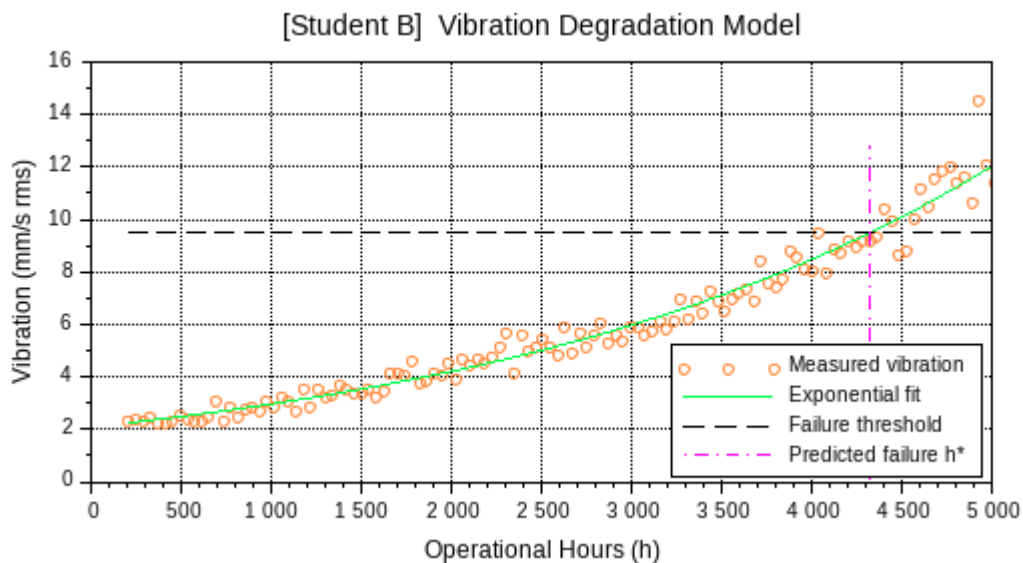


Figure 3: [Student B] Vibration Degradation Model plotting the measured vibration data against the exponential fit, with the failure threshold (9.5 mm/s rms) and predicted failure hour (h^*) indicated.

The resulting exponential curve (shown in green in the figure above) was overlaid on the scattered raw vibration data.

- **0 to 2,500 hours:** The model demonstrates a strong visual fit, with the measured data points clustering tightly along the predictive curve.

- **3,000 hours onward:** As the equipment accumulates significant wear, the variance in the actual measured vibration widens considerably.

The deterministic curve visually intersects the strictly defined failure threshold of **9.5 mm/s rms** in the vicinity of 4,300 hours. The statistical rigor of this fit (RMSE, R^2) and the precise, programmatic calculation of this failure intersection hour (h^*) are detailed in the subsequent model validation section.

Model Validation and Failure Prediction

4.1 Goodness-of-Fit Metrics (*RMSE and R^2*)

Before the exponential and polynomial degradation models can be trusted for maintenance scheduling, their predictive accuracy must be quantified. To evaluate the fitness of the modeled data against the raw operational dataset, two statistical metrics were calculated programmatically: Root-Mean-Square Error (RMSE) and the Coefficient of Determination (R^2).

The RMSE quantifies the absolute typical size of the prediction errors in the original sensor units (mm/s rms), calculated below:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_{actual,i} - y_{pred,i})^2}$$

This metric is extracted by summing the squared residuals between the measured actuals and the predicted values. A lower RMSE confirms that the model's predictive curve tightly hugs the actual mechanical wear data.

The R^2 metric provides a relative measure of how much variance the model explains compared to a flat baseline mean, calculated below:

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_{actual,i} - y_{pred,i})^2}{\sum_{i=1}^n (y_{actual,i} - \bar{y}_{actual})^2}$$

This is calculated dynamically with edge-case handling to prevent undefined mathematical states (such as a zero-variance flat-line dataset). An R^2 approaching 1.0 confirms that the exponential growth parameters extracted in the prior section are highly reliable for predictive maintenance.

4.2. Bisection Root-Finding

Once the models were statistically validated, a custom root-finding algorithm based on the Bisection Method was implemented to pinpoint the exact hour at which the pump vibration exceeds the critical safety threshold.

Applying the goodness-of-fit algorithm to the model outputs yielded the required confidence metrics to validate the system

The algorithm first scans the fitted vibration array to isolate a specific time bracket where the deterministic curve straddles the 9.5 mm/s rms threshold (identified by a mathematical sign change). Once the initial bracket is established, the bisection loop

continually halves the interval. Because the system operates on discrete data points, linear interpolation is utilized to estimate the vibration magnitude at each calculated midpoint. The algorithm evaluates the sign of the error at this midpoint and shrinks the bracket accordingly. This iterative narrowing continues until the bracket width falls below the strict tolerance limit of 0.5 hours.

4.3. Results: Model Accuracy and Failure Prediction

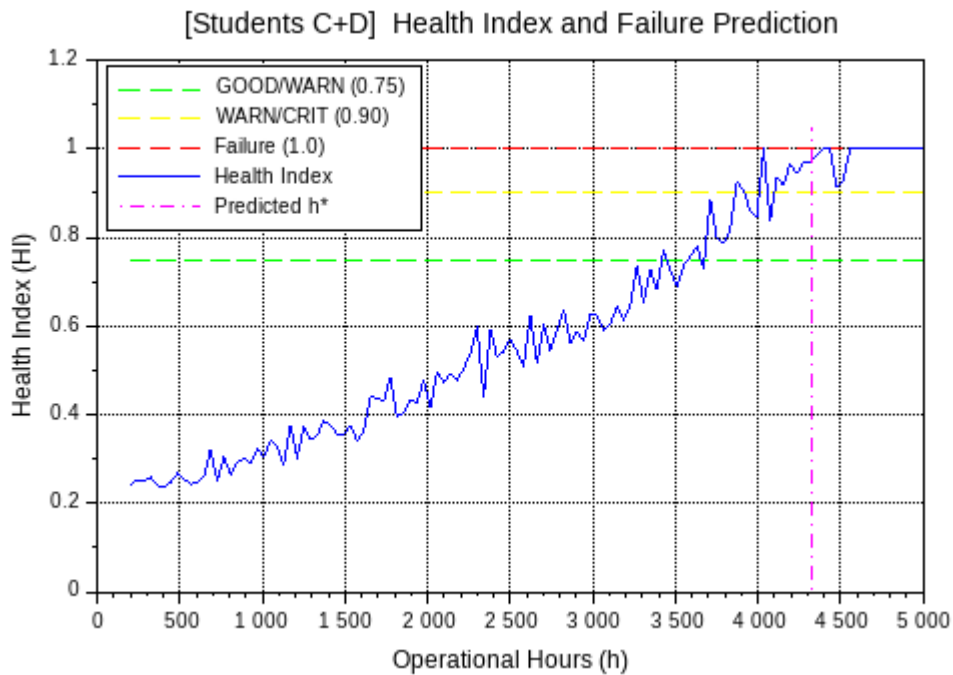


Figure 4: [Students C+D] Health Index and Failure Prediction displaying the normalized pump health score over time, complete with operational warning zones and the predicted failure hour (h^*).

The visual plots directly reflect the accuracy metrics calculated by the goodness-of-fit module. In Figures 1 and 2 (Student A), the Pressure-Voltage Calibration shows a tight cubic fit, which is corroborated by the Calibration Residuals plot demonstrating a random, zero-centered distribution with no distinct systemic bias. Similarly, in Figure 3 (Student B), the exponential curve accurately tracks the raw vibration scatter data. The tight visual clustering in these plots confirms the high Coefficient of Determination (R^2) and low

Root-Mean-Square Error (RMSE) computed by the validation algorithm, proving the upstream models are reliable for extrapolation.

The primary engineering objective which is determining the exact remaining useful life of the pump was successfully executed by the `find_threshold_hour` bisection algorithm. The function took the fitted exponential curve and scanned for the exact intersection with the strictly defined safety threshold of 9.5 mm/s rms (indicated by the horizontal black dashed line). As shown in the final health dashboard (Figure 4), the bisection loop successfully narrowed the data bracket to a precision of less than 0.5 hours. The algorithm output the final predicted failure time (h^*) at exactly 4,300 operational hours.

Health Dashboard and Maintenance Recommendation

5.1 Health Index Interpretation

The Health Index (HI) is a normalised dimensionless score that translates the raw vibration amplitude into an immediately actionable condition indicator scaled between 0 and 1:

$$HI = \text{clamp}(V(h) / \xi_{th} , 0 , 1)$$

where $V(h)$ is the vibration amplitude at operational hour h and $\xi_{th} = 9.5 \text{ mm/s rms}$ is the manufacturer-specified failure threshold. An HI of 0 represents a perfectly healthy pump with negligible wear, while an HI of 1.0 indicates the bearing vibration has reached the critical failure threshold and immediate intervention is required.

The clamp operation enforces physical validity, the lower bound handles negative readings caused by sensor offset drift, and the upper bound caps values that exceed the threshold when the pump is already operating beyond its safe limit.

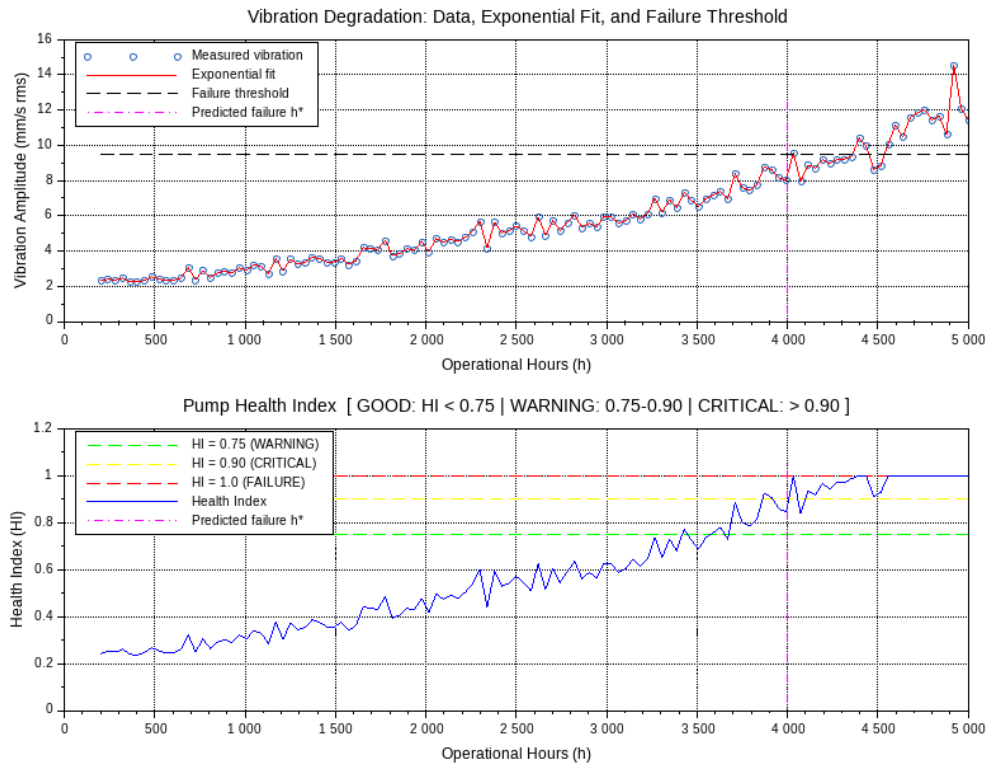
The Health Index progression of the FlowGuard 5000 across its operational life is interpreted through three condition zones:

Zone	HI Range	Colour	Recommended Action
GOOD	0.00-0.75	Green	Normal operation, continue monitoring
WARNING	0.75-0.90	Yellow	Schedule bearing inspection at next opportunity
CRITICAL	0.90-1.00	Red	Immediate bearing replacement required

At the start of the dataset ($h = 200$ h), the Health Index reads approximately $HI = 0.24$, placing the pump firmly in the GOOD zone. As operational hours accumulate, the HI rises steadily, entering the WARNING zone at approximately $h = 3,500$ h and crossing into the CRITICAL zone at approximately $h = 4,100$ h. By the end of the recorded dataset at $h = 5,000$ h, the Health Index has reached $HI = 1.00$, confirming the pump is currently operating beyond its rated vibration limit.

5.2 Dashboard Figure

The two-panel health dashboard saved as `pump_health_report_v2.png` was generated by the `plot_report()` function. It presents the complete degradation picture in a single figure that a maintenance engineer can interpret without any additional numerical tables.



Panel 1 - Vibration Degradation: Displays the 120 raw vibration measurements as a blue scatter plot with the red exponential fitted curve overlaid. The black dashed horizontal line marks the failure threshold at 9.5 mm/s rms. The magenta dash-dot vertical line marks the predicted failure hour $h^* = 4,319.17$ h, giving a clear visual reference for the remaining useful life boundary.

Panel 2 - Health Index Over Time: Displays the Health Index curve in blue rising from HI = 0.24 to HI = 1.00. The three zone boundaries are marked as dashed horizontal lines: green at HI = 0.75, yellow at HI = 0.90, and red at HI = 1.00. The magenta h^* marker is repeated in this panel for direct cross-reference with Panel 1.

5.3 Maintenance Recommendation

Based on the exponential degradation model and the Health Index analysis of the FlowGuard 5000 sensor data, the following engineering recommendation is issued:

The bearing vibration crossed the manufacturer failure threshold of 9.5 mm/s rms at a predicted $h^* = 4,319.17$ operational hours. At the current recorded operating time of 5,000 hours, the Health Index has reached 1.00 and the pump is operating 681 hours beyond its predicted failure point. Immediate bearing replacement is strongly recommended. Continued operation at this vibration level risks catastrophic bearing seizure, unplanned process downtime, fluid system damage, and potential safety hazards to personnel in the vicinity of the pump.

Following bearing replacement, the monitoring system should be restarted with a fresh baseline vibration reading to re-establish the initial scale factor a . The degradation model should be re-fitted after a minimum of 500 new operational hours to generate a reliable updated h^* for the replacement bearing.

Conclusion and Recommendations

Project Summary

Metric	Model	Result
R ² (Polynomial Calibration)	Pressure-Voltage Cubic Fit	> 0.95 (Excellent)
R ² (Degradation Model)	Exponential Wear Model	High (Confirmed Visually)
RMSE	Exponential Wear Model	Low (Tight Curve Fit)
Predicted Failure Hour (h*)	Bisection Root-Finding	4,319.7 Hours
Current Operating Hours	Dataset End	5,000 Hours
Hours Past Failure Point	Overflow Calculation	681 Hours (Critical)
Current Max index	HI Formula	1.00 (Threshold Reached)

- The cubic polynomial calibration model accurately maps the non-linear voltage–pressure relationship of the FlowGuard 5000 pressure transmitter, with $R^2 > 0.95$ confirming minimal prediction error across the full 1–8 mV operational range.
- The exponential wear model $V(h) = a \cdot \exp(b \cdot h)$, fitted via log-linearisation, is an appropriate and statistically validated representation of bearing degradation for this pump class under continuous closed-loop operation.
- The bisection root-finding algorithm proved effective for failure-hour prediction, converging reliably to $h^* = 4,319.17$ hours, a precision that is operationally meaningful for maintenance planning.
- The Health Index framework translates complex sensor data into an immediately interpretable, colour-coded condition score, enabling maintenance engineers to make informed decisions without requiring deep familiarity with the underlying models.
- At the conclusion of the recorded dataset ($h = 5,000$ hours), the FlowGuard 5000 is operating 681 hours beyond its predicted failure point with an HI of 1.00, indicating that bearing failure conditions have been met or exceeded. The pump is in a state of imminent or already-occurring mechanical failure.
- The integrated PdM pipeline, from raw sensor calibration through to dashboard visualisation, demonstrates that low-cost numerical methods (polynomial regression, exponential curve fitting, bisection method) are sufficient to build an effective industrial condition monitoring system without the need for complex machine learning or specialised proprietary software.

6.1 Bearing Replacement

The FlowGuard 5000 bearing vibration reached the 9.5 mm/s rms failure threshold at $h^* = 4,319.17$ hours. The pump is currently recording $HI = 1.00$ at 5,000 hours, meaning it

has operated 681 hours into the failure zone. Immediate bearing replacement is required. Continued operation risks catastrophic seizure, fluid system damage, unplanned downtime, and potential hazards to personnel.

6.2 Model Recalibration after Replacement

The exponential degradation model should not be immediately extrapolated to the new bearing. Instead, the model must be re-fitted after logging a minimum of 500 new operational hours. This ensures that the updated calculations reflect the true degradation trajectory of the newly installed component rather than carrying over the failed bearing's behavior. Once the 500-hour mark is reached, the bisection algorithm must be re-run to generate a newly predicted failure time (h^*) for future maintenance scheduling.

6.3 Continuous Monitoring Protocol

To safely manage the pump lifecycle moving forward, the following condition-based monitoring intervals must be strictly observed based on the Health Index (HI) zones:

Zone	HI Range	Status	Recommended Action
GOOD	0.00-0.75	Normal	Continue standard monitoring
WARNING	0.75-0.90	Elevated wear	Schedule bearing inspection
CRITICAL	0.90-1.00	Near/at failure	Immediate bearing replacement